

Verification 07_H_Simulator_FVM_2D

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1 Objective

The purpose of this verification exercise is to demonstrate that the numerical simulator correctly reproduces the analytical Theis solution for radial slightly compressible single-phase flow in a porous medium.

2 Governing Equation

For two-dimensional slightly compressible single-phase flow in a heterogeneous porous medium, conservation of mass combined with Darcy's law gives

$$\frac{\partial (\phi \rho)}{\partial t} - \frac{1}{\mu} \nabla \cdot (\rho k \nabla P) = \frac{Q}{V}. \quad (1)$$

where

- P is pressure [Pa],
- ρ is fluid density [kg/m³],
- ϕ is porosity [-],
- k is permeability [m²],
- Q represents source terms [kg/s],
- V is the cell volume [m³],
- μ is fluid viscosity [Pa·s].

For a slightly compressible fluid,

$$c_f = \frac{1}{\rho} \frac{\partial \rho}{\partial P}, \quad (2)$$

the density is given by

$$\rho = \rho_0 e^{c_f (P - P_0)}, \quad (3)$$

where ρ_0 is the reference density at pressure P_0 . Similarly, ϕ and k are found by,

$$\phi = \phi_0 e^{c_\phi (P - P_0)}, \quad k = k_0 e^{c_k (P - P_0)}. \quad (4)$$

3 Analytical Solution

The analytical solution for radial flow is given by the Theis solution,

$$P(r, t) = P_0 + \frac{\mu Q}{4\pi \rho k h} E_1 \left(\frac{r^2}{4\alpha t} \right), \quad (5)$$

where E_1 is the exponential integral and r is the radial distance from the well injecting at constant rate.

4 Simulation Parameters

The simulation parameters are given in Table 1.

Parameter	Symbol	Value
Final time	t_f	3000 s
Number of cells	N_x	101
Number of cells	N_y	121
Domain length	L_x	700 m
Domain width	L_y	500 m
Domain height	h	1 m
Initial permeability	k_0	$1 \times 10^{-12} \text{ m}^2$
Viscosity	μ	0.1 Pa·s
Initial porosity	ϕ_0	0.2
Fluid compressibility	c_f	$1 \times 10^{-10} \text{ Pa}^{-1}$
Permeability “compressibility”	c_k	0 Pa ⁻¹
Rock compressibility	c_ϕ	0 Pa ⁻¹
Reference density	ρ_0	1000 kg/m ³
Reference pressure	P_0	$1 \times 10^5 \text{ Pa}$
Well mass rates	Q_w	1 kg/s
Constant mass-rate well locations	x_m	350 m
Constant mass-rate well locations	y_m	250 m

Table 1: Simulation parameters

5 Results

Using the parameters in Table 1, the pressure solutions for both the MATLAB simulator, Figure 1, and the Python simulator, Figure 2, can be seen in 2D. These pressure solutions match the Theis solution for radial flow very well, Figure 3 and Figure 4.

6 Discussion

The numerical pressure distribution reproduces the analytical pressure solution.

7 Conclusion

The simulator successfully reproduces the analytical solution for radial slightly compressible flow. This provides verification that the finite volume implementation is correctly solving the governing equations for this benchmark problem.

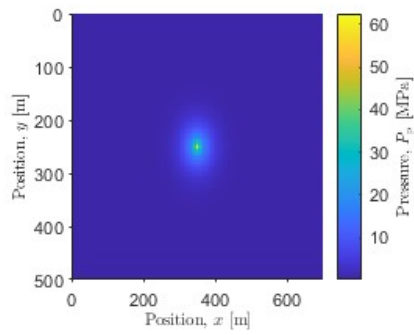


Figure 1: Pressure field in 2D from MATLAB simulation.

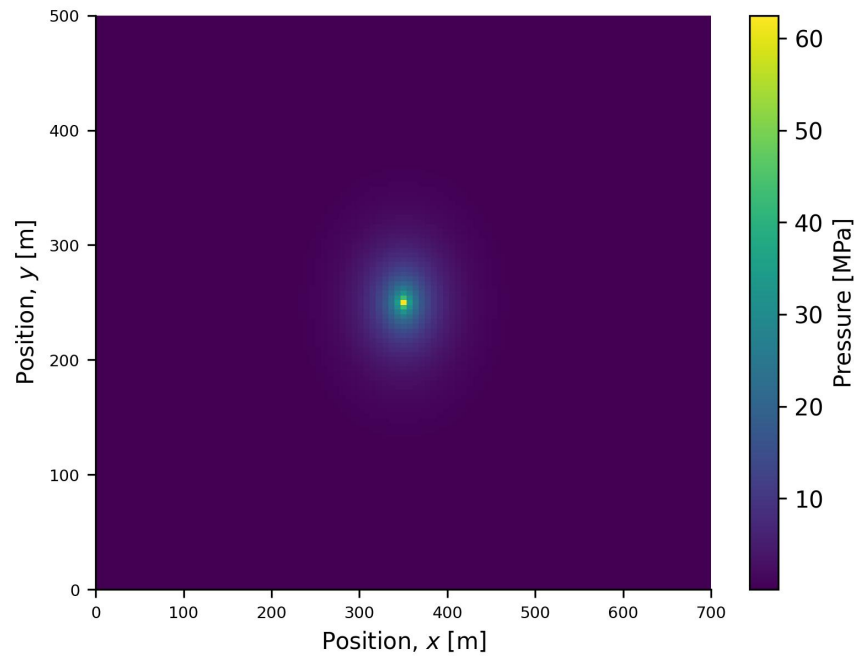


Figure 2: Pressure field in 2D from Python simulation.

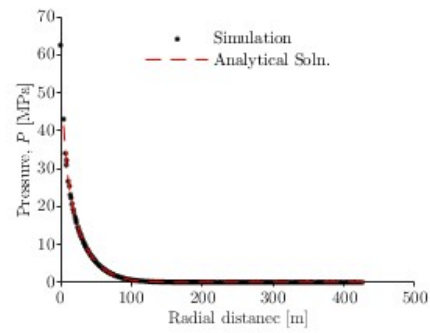


Figure 3: Comparison of the 2D MATLAB simulation with the Theis solution.

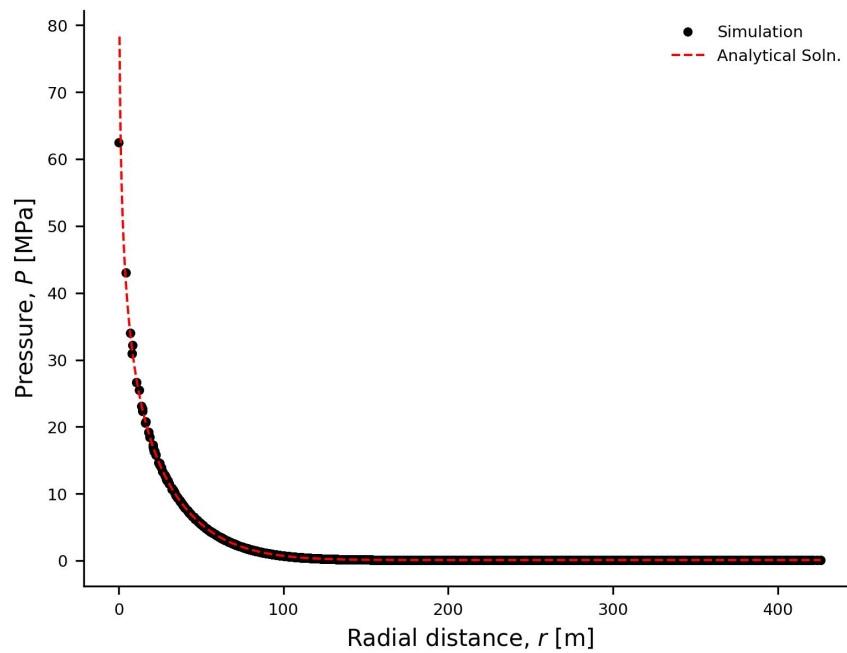


Figure 4: Comparison of the 2D Python simulation with the Theis solution.